CSA0609-DESIGN AND ANALYSIS OF ALGORITHMS

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Exp-11: Travelling Salesperson Problem (TSP) Using Exhaustive Search

Aim:

To find the minimum-cost tour for the Travelling Salesperson Problem using the **Exhaustive Search** technique.

Algorithm:

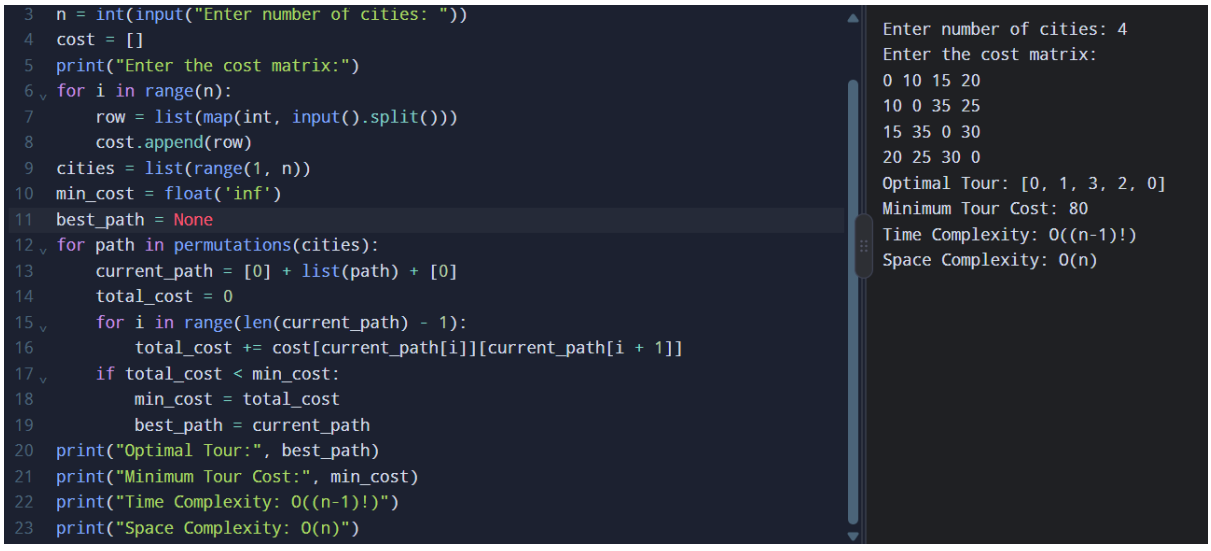
1. Read the number of cities.
2. Read the cost/distance matrix between the cities.
3. Start from city 0.
4. Generate all possible permutations of the remaining cities.
5. For each permutation, calculate the total tour cost.
6. Add the cost of returning to the starting city.
7. Compare all possible tour costs.
8. Select the tour having the minimum cost.
9. Display the optimal tour, minimum cost, and complexity.

Source Code:

```
from itertools import permutations
n = int(input("Enter number of cities: "))
cost = []
print("Enter the cost matrix:")
for i in range(n):
    row = list(map(int, input().split()))
    cost.append(row)
cities = list(range(1, n))
```

```
min_cost = float('inf')
best_path = None
for path in permutations(cities):
    current_path = [0] + list(path) + [0]
    total_cost = 0
    for i in range(len(current_path) - 1):
        total_cost += cost[current_path[i]][current_path[i + 1]]
    if total_cost < min_cost:
        min_cost = total_cost
        best_path = current_path
print("Optimal Tour:", best_path)
print("Minimum Tour Cost:", min_cost)
print("Time Complexity: O((n-1)!)")
print("Space Complexity: O(n)")
```

Output:



Result:

The minimum-cost tour for the Travelling Salesperson Problem was successfully obtained using the Exhaustive Search technique.